

Digital Logic Design - Theory [Assignment #2]

Name:-

Sana Munir Alam

Roll #:

24K-0573

Class:-

BSCS-2E

Q1. Write the boolean expression for the following logic circuits and simplify it.

a) $Y = (\overline{A}B) \cdot \overline{C} \cdot B$ } actual Boolean expression

Simplifying

$$(\overline{A} + \overline{B}) \overline{C} B$$

$$\overline{A} B \overline{C} + B \overline{B} \overline{C}$$

$$\overline{A} B \overline{C}$$

$$\therefore Y = \overline{A} B \overline{C} \quad \text{ } \} \text{ simplified Boolean Expression}$$

b) $Z = \underbrace{\underbrace{(B + \overline{C})}_P(A)}_Q + \underbrace{\underbrace{(C + \overline{D})}_R(B)}_S + \underbrace{(BD)}_T$ } actual boolean expression

Simplifying

$$A(B + \overline{C}) + B(C + \overline{D}) + BD$$

$$AB + A\overline{C} + BC + B\overline{D} + BD$$

$$AB + A\overline{C} + BC + B(\overline{D} + D)$$

$$AB + A\overline{C} + BC + B$$

$$B(A + 1) + A\overline{C} + BC$$

$$B + A\overline{C} + BC$$

$$B(1 + C) + A\overline{C}$$

$$B + A\overline{C}$$

$$\therefore Z = B + A\overline{C} \quad \text{ } \} \text{ simplified Boolean expression}$$

c) $F = (X + \bar{Y})(Z) + (Y\bar{X}\bar{Z})$ } actual Boolean expression

Simplifying

$$Z(X + \bar{Y}) + (Y\bar{X}\bar{Z})$$

$$ZX + Z\bar{Y} + Y\bar{X}\bar{Z}$$

can't be further simplified hence actual expression is same for simplified expression

$\therefore F = Z(X + \bar{Y}) + (Y\bar{X}\bar{Z})$ } simplified Boolean expression.

OR $F = ZX + Z\bar{Y} + Y\bar{X}\bar{Z}$ both are same thing

d) $Y = \overline{A(B \cdot \bar{B}C)} + ((B \cdot \bar{B}C) \oplus (\bar{B}C))$ } actual boolean expression

Simplifying

$$\underbrace{\overline{AB\bar{B}C}}_{LHS} + \underbrace{(B\bar{B}C \oplus \bar{B}C)}_{RHS}$$

simplifying LHS

$$\overline{B\bar{A}BC}$$

~~Let~~ Let $\bar{A}BC = X$

so $\overline{BX}$

applying De morgan

$$\overline{B} + \overline{X}$$

$$\overline{B} + \overline{\bar{A}BC}$$

applying De morgan

$$\overline{B} + \overline{\bar{A}} + \overline{BC}$$

$$A + \bar{B} + \bar{C}$$

applying Absorption Law

$$A + \bar{B} + C$$

simplifying RHS

$$B\bar{B}C \oplus \bar{B}C$$

$$B\bar{B}C \oplus BC$$

applying De morgan

$$B(\bar{B} + \bar{C}) \oplus BC$$

$$(B\bar{B} + B\bar{C}) \oplus BC$$

$$B\bar{C} \oplus BC$$

$$\therefore Y = A + \bar{B} + C + (B\bar{C} \oplus BC)$$

simplified Boolean expression



However if we further simplify $B\bar{C} \oplus BC$ then

$$\text{Let } X = B\bar{C} \text{ and } Y = BC$$

$$\text{so } X \oplus Y = X\bar{Y} + \bar{X}Y$$

$$(B\bar{C})(\overline{BC}) + (\overline{B\bar{C}})(BC)$$

$$(B\bar{C})(\bar{B} + \bar{C}) + (BC)(\bar{B} + C)$$

$$B\bar{B}\bar{C} + B\bar{C}\bar{C} + B\bar{B}C + BCC$$

$$0 + B\bar{C} + 0 + BC$$

$$B\bar{C} + BC$$

$$B(\bar{C} + C)$$

$$B$$

$$\therefore A + \bar{B} + C + (B\bar{C} \oplus BC)$$

can be written as

$$A + \bar{B} + C + B$$

further simplify

$$= A + C + (B + \bar{B})$$

$$= A + (C + 1)$$

$$= A + 1$$

$$= 1$$

$\therefore Y = 1$ the most simplified expression.

Q2) $\overline{A}BC + A\overline{B}C + \overline{A}B\overline{C} + A\overline{B}\overline{C} + ABC$

now there is a problem in this question, if we simplify it like this we get 1 as the simplest form, therefore ~~the~~ ^{simplified} truth table and circuit can't be made.

Let me prove

Original Truth table

A	B	C	$\overline{A}$	$\overline{B}$	$\overline{C}$	$\overline{A}B\overline{C}$	$A\overline{B}C$	$\overline{A}B\overline{C}$	$A\overline{B}C$	$\overline{A}B\overline{C}$	$A\overline{B}C$	X
0	0	0	1	1	1	0	0	1	0	0	0	1
0	0	1	1	1	0	0	0	1	0	0	0	1
0	1	0	1	0	1	0	0	1	0	0	0	1
0	1	1	1	0	0	1	0	1	0	0	0	1
1	0	0	0	1	1	0	1	1	0	0	0	1
1	0	1	0	1	0	0	1	1	1	0	0	1
1	1	0	0	0	1	0	1	1	0	0	0	1
1	1	1	0	0	0	0	0	0	0	1	1	1

INPUT output

Simplify

$$\overline{A}BC + A\overline{B}C + \overline{A}B\overline{C} + A\overline{B}\overline{C} + ABC$$

$$\overline{A}BC + A(\overline{B} + \overline{C}) + (\overline{A} + \overline{B} + \overline{C}) + A\overline{B}C + ABC \quad \text{\{ De-morgan theorem \}}$$

$$\overline{A} + \overline{B} + \overline{C} + A\overline{B} + A\overline{C} + \overline{A}B\overline{C} + A\overline{B}C + ABC$$

$$\overline{A}(1 + BC) + \overline{B}(1 + AC) + \overline{C} + A\overline{B} + A\overline{C} + ABC$$

$$\overline{A} + \overline{B} + \overline{C} + A\overline{B} + A\overline{C} + ABC$$

$$\overline{A} + \overline{B}(1 + A) + \overline{C}(1 + A) + ABC$$

$$\overline{A} + \overline{B} + \overline{C} + ABC$$

$$\overline{A} + A(BC) + \overline{B} + \overline{C} \quad \text{\{ rule 15b \}}$$

$$\overline{A} + \overline{B} + \overline{C} + BC$$

$$\overline{A} + \overline{B} + B(C) + \overline{C} \quad \text{\{ rule 15b \}}$$

$$\overline{A} + \overline{B} + C + \overline{C} \quad \text{\{ rule 8 now applied \}}$$

$$\overline{A} + \overline{B} + 1$$

$$1$$

Note but if we are to make truth table and circuit than we can make it from this expression

$$\overline{A} + \overline{B} + \overline{C} + BC$$

see << page LHS

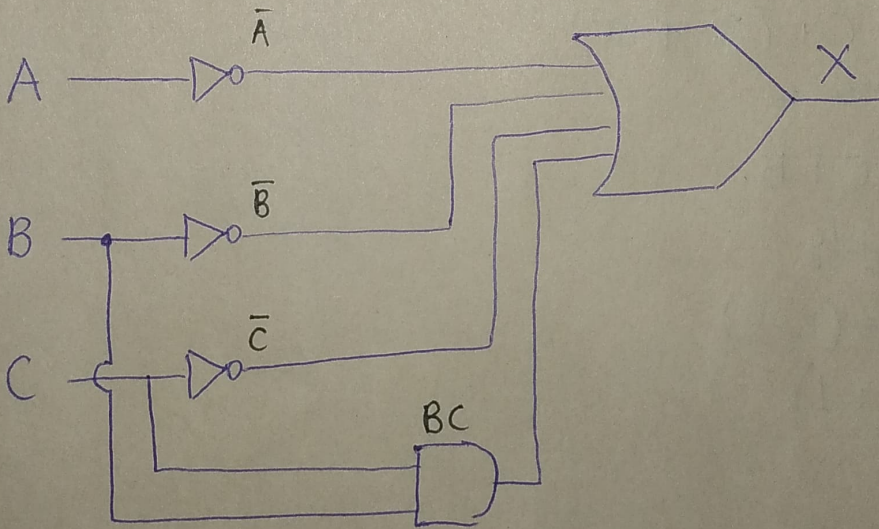
for truth table and circuit

Q2) Making truth table of expression simplified

$$X = \bar{A} + \bar{B} + \bar{C} + BC$$

A	B	C	$\bar{A}$	$\bar{B}$	$\bar{C}$	BC	X
0	0	0	1	1	1	0	1
0	0	1	1	1	0	0	1
0	1	0	1	0	1	0	1
0	1	1	1	0	0	1	1
1	0	0	0	1	1	0	1
1	0	1	0	1	0	0	1
1	1	0	0	0	1	0	1
1	1	1	0	0	0	1	1

Logic circuit



Solution for Q2 problem (version 2) if applicable else ignore

However we can indeed make truth table and circuit of simplified form if we consider each complement separately.

$$\bar{A}BC + A\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C} + A\bar{B}C + ABC$$

simplify

$$\bar{B}\bar{C}(\bar{A}+A) + \bar{A}BC + AC(\bar{B}+B)$$

$$\bar{B}\bar{C} + ABC + AC$$

$$\bar{B}\bar{C} + C(A + \bar{A}B)$$

$$\bar{B}\bar{C} + C(A + B)$$

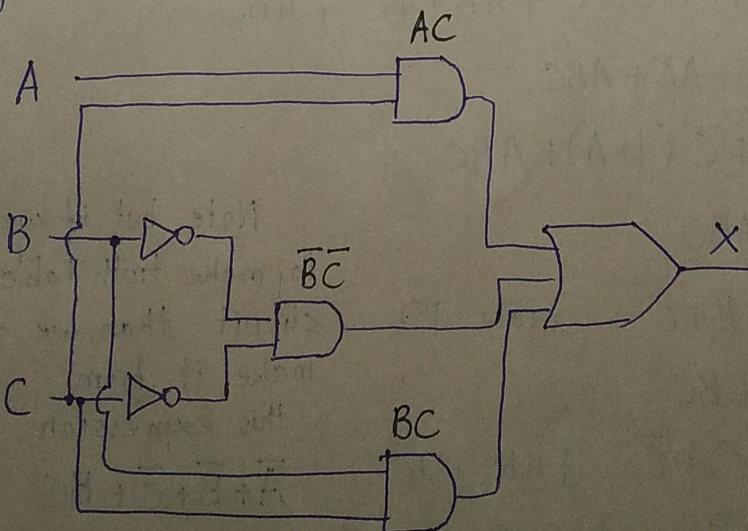
$$AC + BC + \bar{B}\bar{C} \quad \} \text{ simplified form}$$

A	B	C	$\bar{B}$	$\bar{C}$	AC	BC	$\bar{B}\bar{C}$	X
0	0	0	1	1	0	0	1	1
0	0	1	1	0	0	0	0	0
0	1	0	0	1	0	0	0	0
0	1	1	0	0	0	1	0	1
1	0	0	1	1	0	0	1	1
1	0	1	1	0	1	0	0	1
1	1	0	0	1	0	0	0	0
1	1	1	0	0	1	1	0	1

Simplified Boolean
Expressions
Truth Table

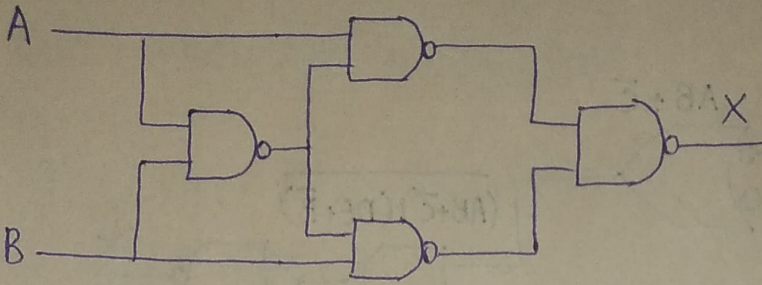
(Note:- The value of
simplified truth table
matched the above
expression truth table)

Logic Circuit.

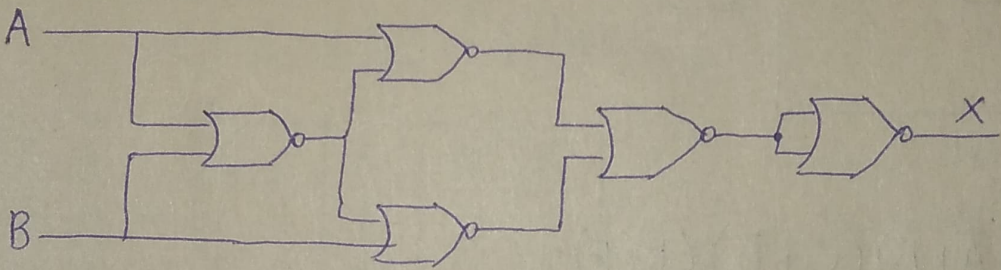


Q3i) $\bar{A}B + \bar{B}A$

Logic circuit using NAND

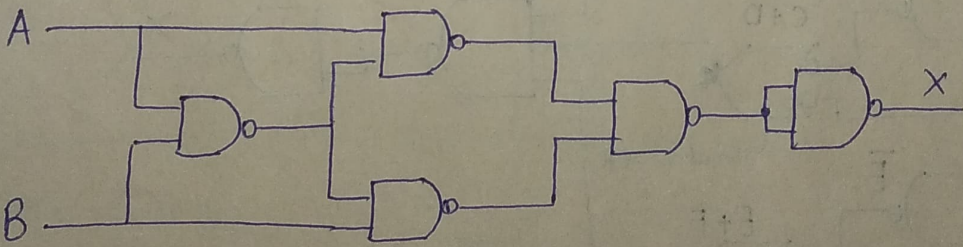


Logic Circuit using NOR

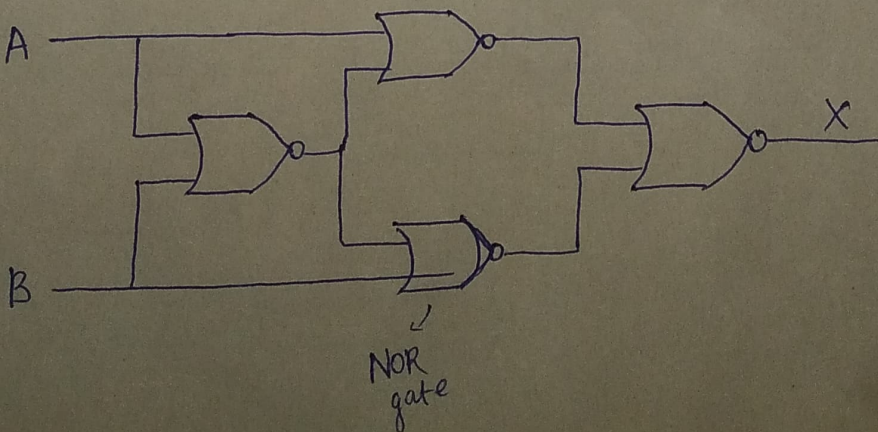


Q3ii) $\bar{A}\bar{B} + AB$

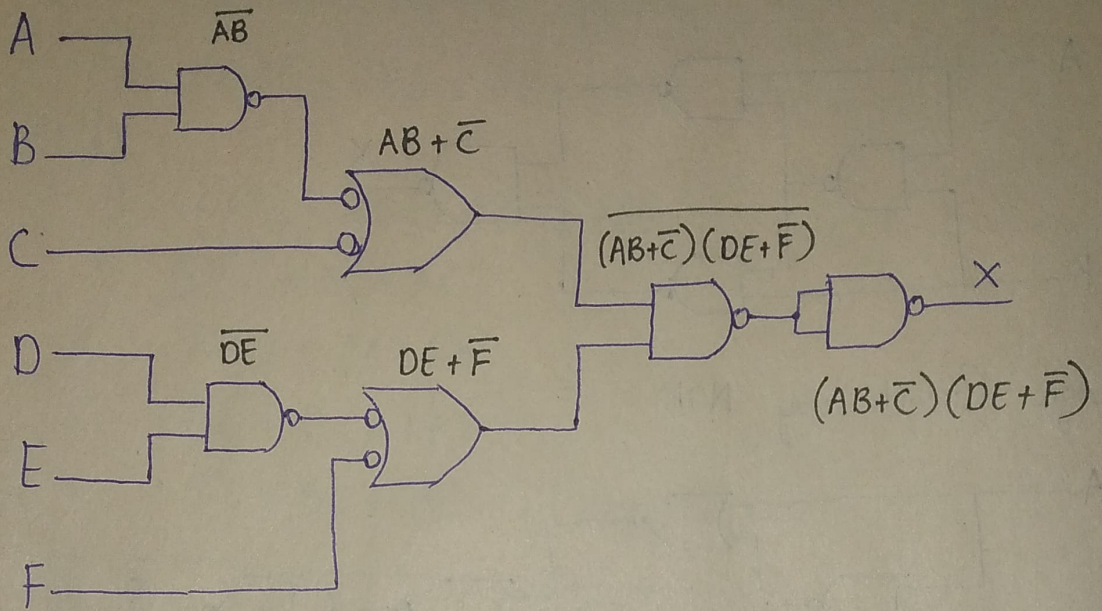
Logic circuit using NAND



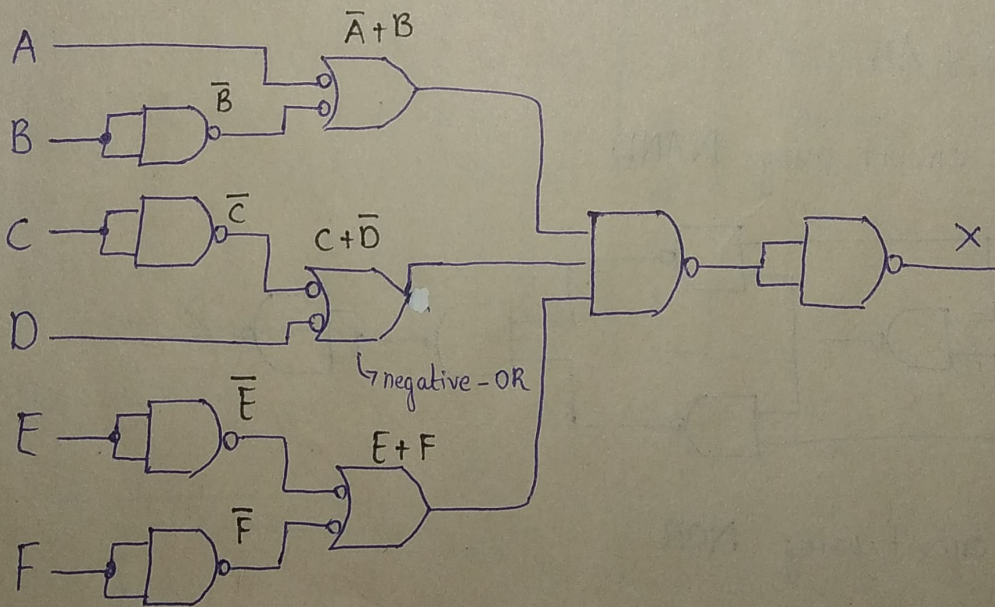
Logic circuit using NOR



Q4i) $(AB + \bar{C})(DE + \bar{F})$



Q4ii) $(\bar{A} + B)(C + \bar{D})(E + F)$



Q5) $F(A, B, C, D) = \sum (0, 1, 2, 8, 9, 12)$

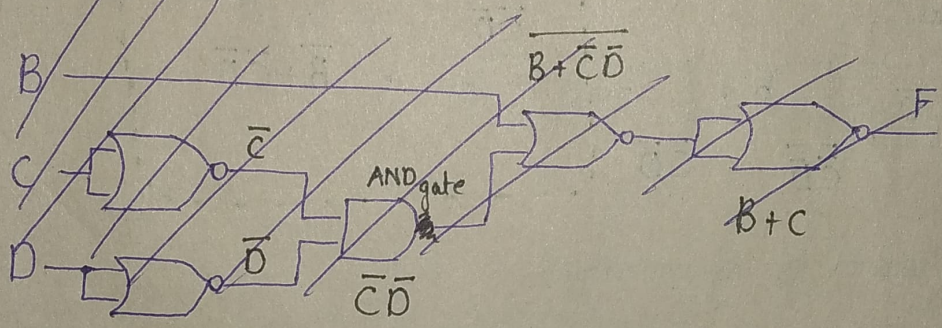
$d(A, B, C, D) = \sum (3, 4, 10, 11)$

	CD			
AB	00	01	11	10
00	1	1	X	1
01	X			
11	1			
10	1	1	X	X

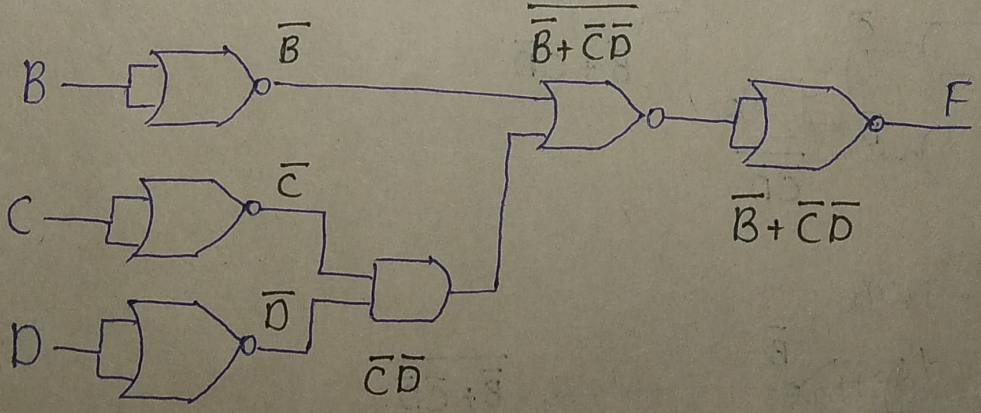
Reduced form

$F = \overline{B} + \overline{C}\overline{D}$

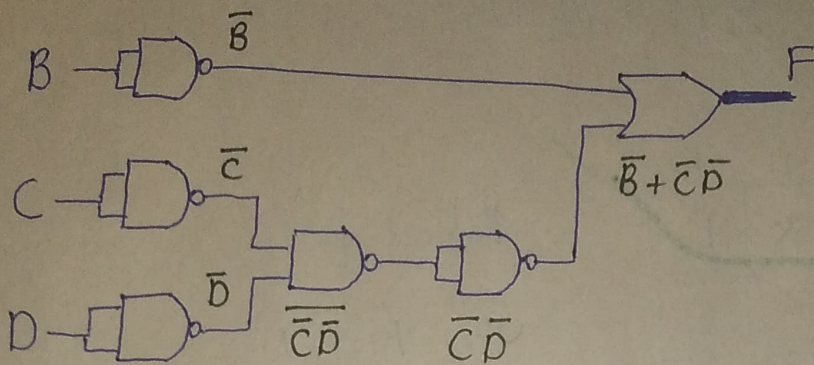
a) AND-NOR



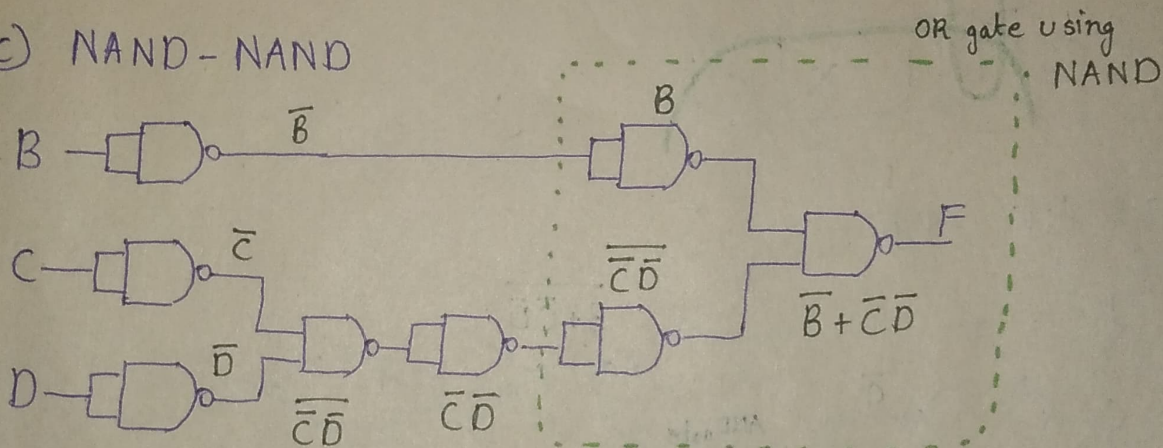
a) AND-NOR



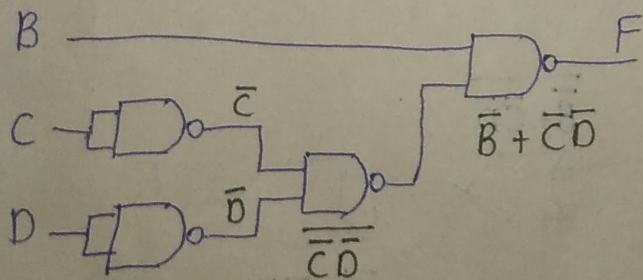
Q56) OR-NAND



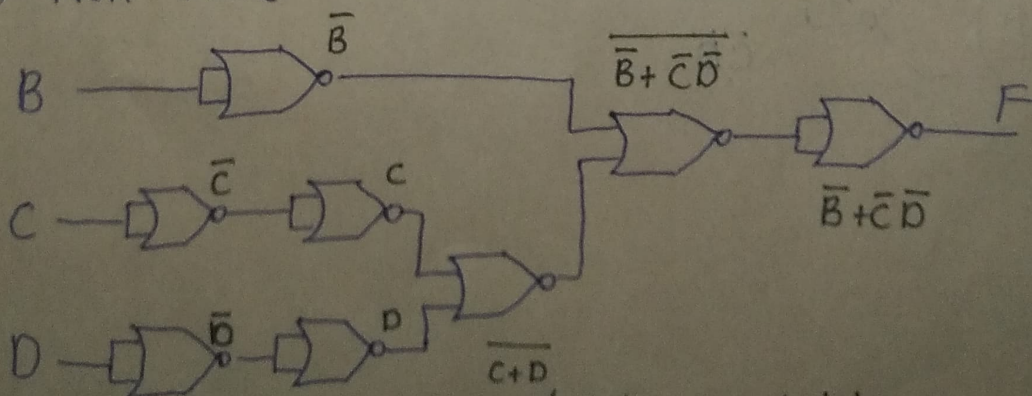
c) NAND-NAND



Simplified version of above circuit



d) NOR-NOR



Which is equivalent to

$\overline{B + C D}$ based upon De-morgan theorem

Q6) $X = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}B\bar{C}$

simplify

$$\bar{A}\bar{B}(\bar{C} + C) + \bar{A}B\bar{C}$$

$$\bar{A}\bar{B} + \bar{A}B\bar{C}$$

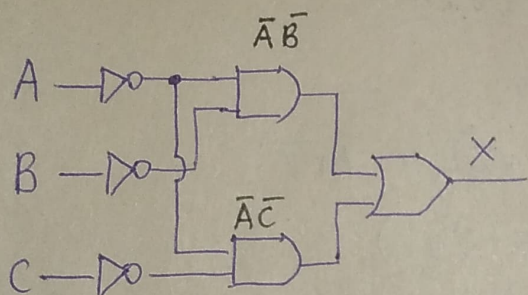
$$\bar{A}(\bar{B} + B\bar{C})$$

} Distributive law

$$\bar{A}(\bar{B} + \bar{C})$$

} absorption law

$$\bar{A}\bar{B} + \bar{A}\bar{C}$$



	A	B	C	$\bar{A}$	$\bar{B}$	$\bar{C}$	$\bar{A}\bar{B}$	$\bar{A}\bar{C}$	X
0	0	0	0	1	1	1	1	1	1
1	0	0	1	1	1	0	1	0	1
2	0	1	0	1	0	1	0	1	1
3	0	1	1	1	0	0	0	0	0
4	1	0	0	0	1	1	0	0	0
5	1	0	1	0	1	0	0	0	0
6	1	1	0	0	0	1	0	0	0
7	1	1	1	0	0	0	0	0	0

Q7)

AB \ CD	00	01	11	10
00	1	0	0	0
01	0	1	1	1
11	0	1	1	1
10	1	0	1	1

Actual Expression:-

$$(\bar{A} + \bar{B} + C + D) \cdot (A + \bar{B} + C + D) \cdot (A + B + C + \bar{D}) \cdot (A + B + \bar{C} + \bar{D}) \cdot (\bar{A} + B + C + \bar{D}) \cdot (A + B + \bar{C} + D)$$

i) Minimum POS expression:

$$(A + B + \bar{C}) \cdot (\bar{B} + C + D) \cdot (B + C + \bar{D})$$

ii) Standard SOP expression:

$$\bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}D + \bar{A}B\bar{C}\bar{D} + \bar{A}B\bar{C}D + A\bar{B}\bar{C}\bar{D} + A\bar{B}\bar{C}D + A\bar{B}C\bar{D} + A\bar{B}CD + AB\bar{C}\bar{D} + ABC\bar{D} + ABCD$$

iii) Minimum SOP expression:-

$$BC + AC + BD + \bar{B}\bar{C}\bar{D}$$

Q8a) $(A+B+C)(\bar{A}+\bar{B}+\bar{C})(A+\bar{B}+C)$

AB \ C	0	1
00	0	
01	0	
11		0
10		

Reduced expression:-

$$(A+C) \cdot (\bar{A}+\bar{B}+\bar{C})$$

b) $(X+\bar{Y})(\bar{X}+Z)(X+\bar{Y}+\bar{Z})(\bar{X}+\bar{Y}+Z)$

first converting it to standard form

$$(X+\bar{Y}) + (\cancel{XZ}) \cdot (\cancel{Z\bar{X}}) = (X+\bar{Y}+Z)(X+\bar{Y}+\bar{Z})$$

$$(\bar{X}+Z) + (\cancel{Y\bar{Y}}) = (\bar{X}+Y+Z)(\bar{X}+\bar{Y}+Z)$$

$$\therefore (X+\bar{Y}+Z)(X+\bar{Y}+\bar{Z})(\bar{X}+Y+Z)(\bar{X}+\bar{Y}+Z)$$

XY \ Z	0	1
00		
01	0	0
11	0	
10	0	

Reduced expression:-

$$(X+\bar{Y}) \cdot (\bar{X}+Z)$$

Q9a) $(A+\bar{B})(A+\bar{C})(\bar{A}+\bar{B}+C)$

first converting it to standard form

$$(A+\bar{B}) + (C\bar{C}) = (A+\bar{B}+C) \cdot (A+\bar{B}+\bar{C})$$

$$(A+\bar{C}) + (B\bar{B}) = (A+B+\bar{C}) \cdot (A+\bar{B}+\bar{C})$$

$$\therefore (A+\bar{B}+C)(A+\bar{B}+\bar{C})(A+B+\bar{C})(\bar{A}+\bar{B}+C)$$

AB \ C	0	1
00	1	0
01	0	0
11	0	1
10	1	1

Reduced Expression of SOP:-

$$\overline{B}\overline{C} + AC$$

b) $(\overline{A}+B)(\overline{A}+\overline{B}+\overline{C})(B+\overline{C}+D)(A+\overline{B}+\overline{C}+\overline{D})$

first converting it into standard form

$$\Rightarrow (\overline{A}+B+C\overline{C}) = (\overline{A}+B+C)(\overline{A}+B+\overline{C})$$

now further standardize each

$$-(\overline{A}+B+C+D\overline{D}) = (\overline{A}+B+C+D)(\overline{A}+B+C+\overline{D})$$

$$-(\overline{A}+B+\overline{C}+D\overline{D}) = (\overline{A}+B+\overline{C}+D)(\overline{A}+B+\overline{C}+\overline{D})$$

$$\Rightarrow (\overline{A}+\overline{B}+\overline{C}+D\overline{D}) = (\overline{A}+\overline{B}+\overline{C}+D)(\overline{A}+\overline{B}+\overline{C}+\overline{D})$$

$$\Rightarrow (B+\overline{C}+D+A\overline{A}) = (A+B+\overline{C}+D)(\overline{A}+B+\overline{C}+\overline{D})$$

same

$$\therefore (\overline{A}+B+C+D) \cdot (\overline{A}+B+C+\overline{D}) \cdot (\overline{A}+B+\overline{C}+D) \cdot (\overline{A}+B+\overline{C}+\overline{D})$$

$$\cdot (\overline{A}+\overline{B}+\overline{C}+D) \cdot (\overline{A}+\overline{B}+\overline{C}+\overline{D}) \cdot (A+B+\overline{C}+D) \cdot (A+\overline{B}+C+\overline{D})$$

AB \ CD	00	01	11	10
00	1	1	1	0
01	1	0	1	1
11	1	1	0	0
10	0	0	0	0

Reduced Expression:-

$$\overline{A}\overline{B}\overline{C} + A\overline{B}\overline{C} + \overline{A}CD + \overline{A}B\overline{D}$$

Q10a) $F(A, B, C, D) = \sum (0, 6, 8, 13, 14)$

$d(A, B, C, D) = \sum (2, 4, 10)$

AB \ CD	00	01	11	10
00	1	0	0	X
01	X	0	0	1
11	0	1	0	1
10	1	0	0	X

orange - POS grp
green - SOP grp

minimum POS :- $(A + \bar{D})(B + \bar{D})(\bar{C} + \bar{D})(\bar{B} + C + D)$

minimum SOP :- $\bar{B}\bar{D} + C\bar{D} + ABC\bar{D}$

Q10b) $F(A, B, C, D) = \sum (1, 3, 5, 7, 9, 15)$

$d(A, B, C, D) = \sum (4, 6, 12, 13)$

AB \ CD	00	01	11	10
00	0	1	1	0
01	X	1	1	X
11	X	X	1	0
10	0	1	0	0

Orange - POS
Green - SOP

AB \ CD	00	01	11	10
00	0	1	1	0
01	X	1	1	X
11	X	X	1	0
10	0	1	0	0

minimum POS :- $(D) \cdot (\bar{A} + B + \bar{C})$

minimum SOP :- $\bar{C}D + BD + \bar{A}D$